جامعة الأميرة نورة بنت عبدالرحمن
Princess Nourah bint Abdulrahman University

DRIVER DROWSINESS DETECTION

Graduation Project

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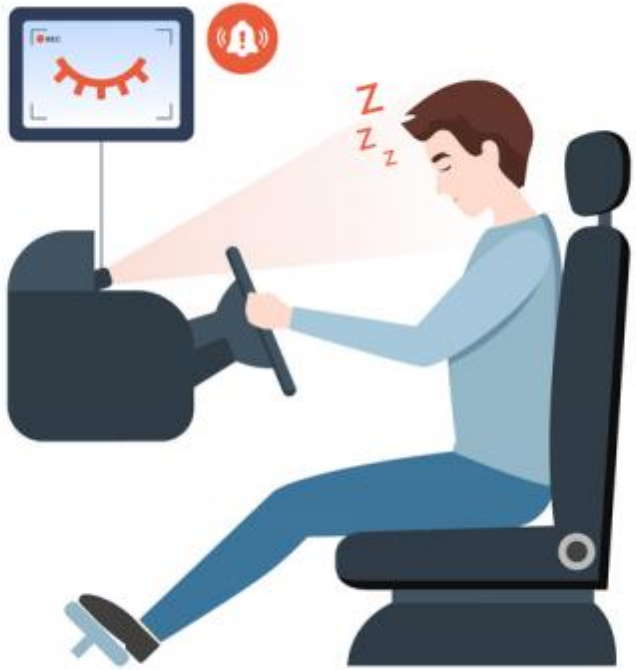
Introduction

- Globally, driver drowsiness is one of the leading causes of traffic accidents, causing a significant risk to drivers and passengers.
- Each year, countless accidents occur due to drowsiness which limits reaction times, focus, and decision-making caused by fatigue.
- Despite advancements in road safety, drowsy driving often goes unnoticed until it’s too late, leading to severe consequences.
- As a result, there is a growing need for innovative and real-time solutions to detect and prevent driver fatigue.
- The development of advanced monitoring systems offers promising potential to address this critical road safety issue.

Problem Statement

- Once a driver becomes drowsy, their ability to react quickly, stay focused, and make good decisions is greatly reduced, increasing the risk of losing control of the vehicle.
- In Saudi Arabia, 20% of drivers reported falling asleep while driving, and 10% have been involved in accidents due to drowsiness.
- There is a critical need for simple, effective systems that can monitor driver alertness in real time and prevent accidents.



Proposed Solution & objectives

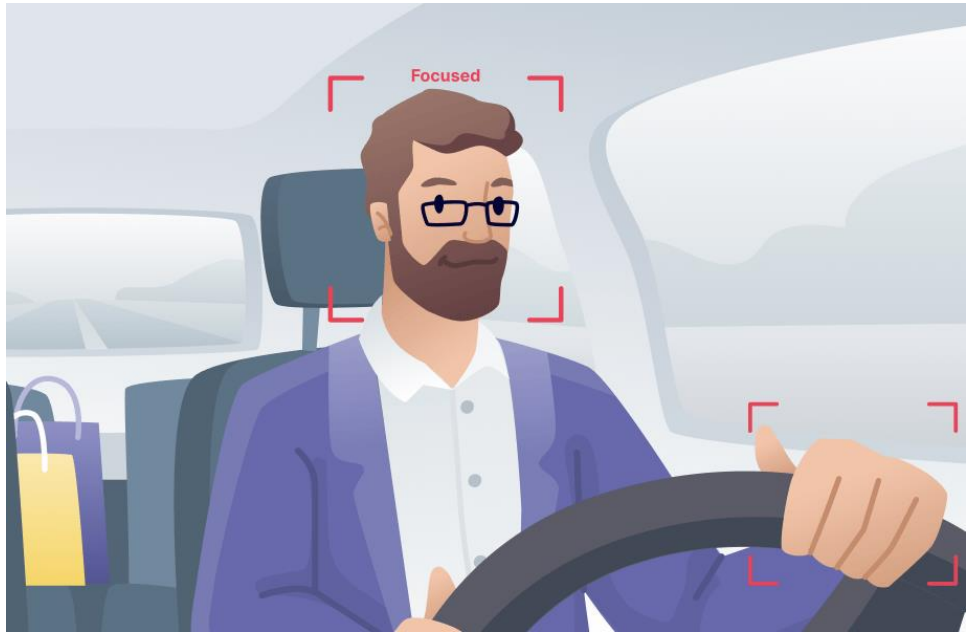
- The integration of AI into the driver drowsiness detection system improves road safety by offering real-time monitoring of drivers.
- The system uses the high-resolution Nicla Vision camera to accurately detect drowsiness signs like eye closure.
- When drowsiness is detected, the system triggers immediate alerts, such as sounds , lights prompting the driver to take corrective action.
- This cost-effective and compact design makes it accessible, providing a reliable solution to reduce accidents caused by fatigue.

Aims & Goals

- Develop a prototype for a driver drowsiness detection system using Arduino, and Nicla vision camera.
- Ensure driver safety by reducing accidents caused by drowsiness.
- Alert the driver immediately with sound or light when drowsiness is detected.
- Use AI to accurately detect signs of drowsiness .

Limitations & Challenges

- **Data Limitations:** Needs a diverse dataset for accurate detection.
- **Installation:** May require professional help, increasing costs.
- **Image Quality:** Relies on clear, high-quality images for effectiveness.



Background Information and Related Work

Artificial intelligence (AI)

- **Artificial Intelligence (AI)** is a broad field aimed at creating machines that perform tasks requiring intelligence, such as learning, problem-solving, and decision-making.
- **Goal is** To create systems that replicate human cognitive abilities and improve over time.

Machine Learning

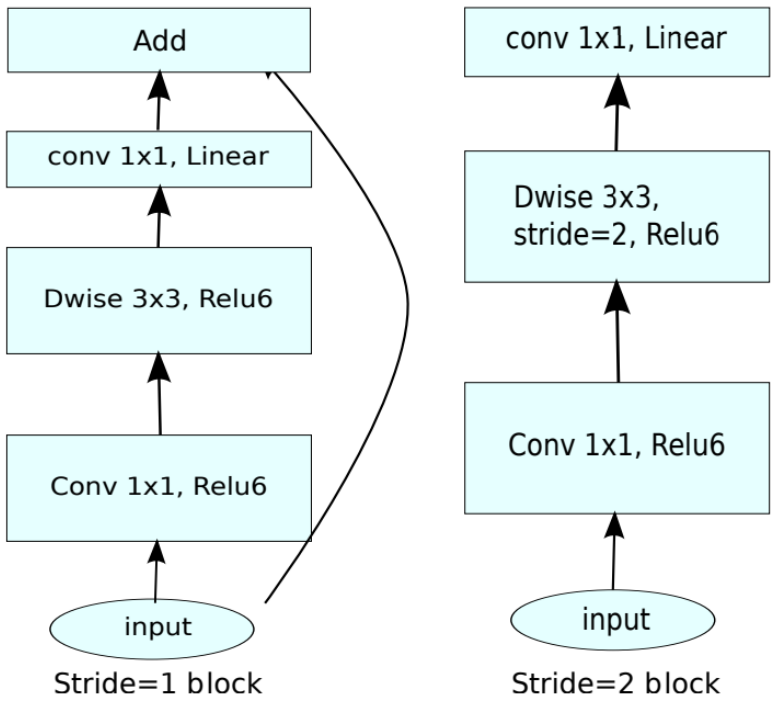
- **Machine Learning** a subset of AI, allows systems to learn from data and improve over time.
- **Supervised Learning:** Trains models on labeled data to predict specific outcomes.
- **Unsupervised Learning**: Deals with unlabeled data to discover hidden patterns.
- **Reinforcement Learning:** Learns through trial and error.

Deep Learning

- **Deep learning** a subfield of ML, uses algorithms inspired by the brain's structure to learn from raw data without manual feature selection.
- **Core Technology**: Convolutional Neural Networks (CNNs) for image-related tasks (classification, object detection).

MobileNet

- **MobileNet** is a lightweight CNN family by Google, optimized for mobile and edge devices with limited computational power.
- Introduced linear bottlenecks for dimension reduction while retaining features.
- enhancing efficiency and accuracy.
- **Applications:**
  - Object detection.
  - Facial recognition .



Edge Computing

- **Edge computing** is a way of processing data closer to where it is created, like on sensors or IoT devices, instead of sending all the data to a central cloud server. This helps to make things faster and more efficient.

- Enables local computation on the "edge" of the network, reducing the need to send data to centralized servers.

Arduino Uno

- **Arduino Uno:** Open-source electronics platform.
- **Known for simplicity, and ease of use.**
- **Supports interactive project prototyping.**
- **Programmed using Arduino IDE with various tools.**

Arduino Nicla Vision Camera

- Arduino Nicla Vision: Compact, standalone camera.
- Analyzes and processes images at the edge.
- Features a 2 MP color camera for high-quality image capture.
- Used for detecting driver drowsiness by monitoring eye blinking

Related work

An Intelligent System for Preventing Accidents Due to Driver distraction

- Monitors and alerts drivers in real-time for distractions.
- Detects drowsiness, phone usage, and head pose using deep learning models.
- **Similarity:** Uses Arduino Uno for system integration and drowsiness detection and Provides audible alerts .
- **Difference:** This system uses Nicla Vision, which is better suited for vehicle environments with on-device processing and compact design.

Related work

Real-Time Driver Drowsiness Detection using Computer Vision

- Detects driver drowsiness and fatigue in real-time to reduce accidents, Uses deep learning models .
- **Similarity :**Both systems aim to detect drowsiness and prevent accidents using real-time monitoring and audio alerts..
- **Difference:** Our project uses Arduino and Nicla Vision for cost-efficient, compact integration, while this system uses Raspberry Pi for higher processing power.

Related work

Evaluation of driver fatigue on two channels of EEG data

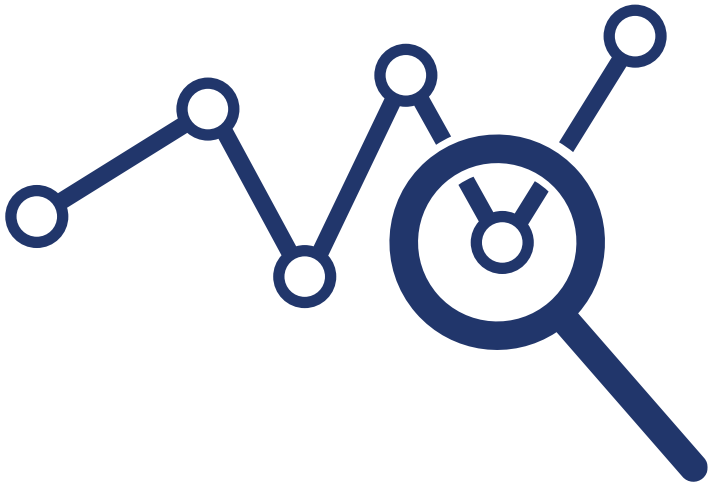
- Detects driver drowsiness by analyzing EEG signals and video inputs , uses EEG headset, Logitech camera.
- **Similarity :**Both systems focus on real-time monitoring and , Utilize cameras to track facial expressions and eye movement.
- **Difference:** Our project is more cost-efficient and lightweight, while EEG-based systems prioritize physiological data for higher accuracy.

Related work

Driver Drowsiness Detection

- Detects driver drowsiness in real-time using computer vision to analyze facial features, Tracking eye closure, yawning, and head nodding
- **Similarity:** Both system use Cameras to monitor facial features, focusing on eye closure as a key indicator.
- **Difference:** Our project prioritize self-sufficiency with embedded systems, while the research system requires additional external processing hardware.

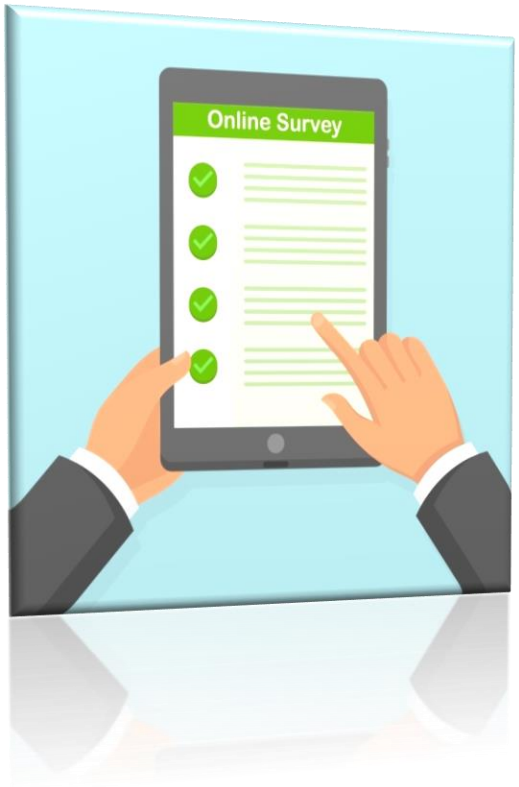
System Analysis



Requirement’s specification

- **Goal**: Develop a prototype to reduce accidents caused by drowsy driving.
- **Technology**: Utilize Deep Learning (DL) to detect signs of driver drowsiness and provide real-time alerts to driver.
- **Hardware**: Arduino Uno (microcontroller) and Nicla Vision.
- **Software**: Arduino IDE and Edge Impulse.
- **Survey**: Conducted with Princess Nora bint Abdul Rahman University (PNU) students who are licensed drivers to specify project requirements.

Survey



- **Purpose**: Understand driver needs to design a helpful drowsiness detection system.
- **Participants**: PNU students with valid driving licenses.
- **Details**: 10 questions about driving habits, drowsiness, and system preferences.

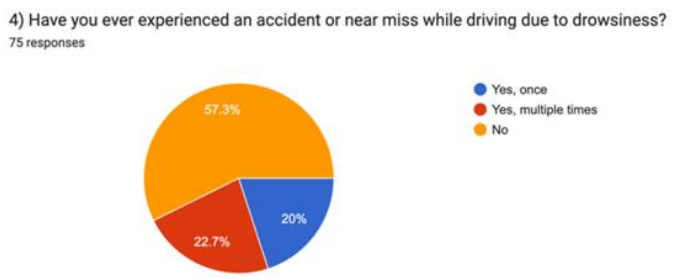
Key topics:

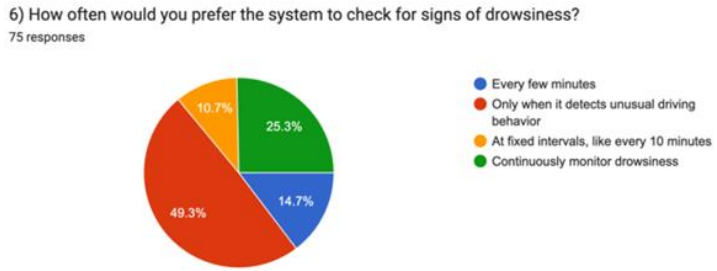
- How often do they drive long distances without breaks.
- Are they Interest in a drowsiness detection system
- how they prefer to be alerted.
- How often do they want the system to check for drowsiness.
- Concerns about false alarms and trust in the system.
- How important they think this system important for road safety.

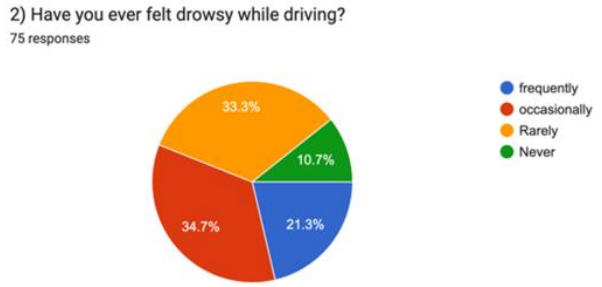
- **Findings**: The survey helped us choose the best hardware and software for the project.

Survey Results Summary

- **Drowsiness Frequency**: Over 50% of drivers have felt drowsy while driving; 22.7% experienced accidents or near-misses.
- **Safety Concerns**: 68% feel safer with a detection system; most want immediate alerts for signs like eye closure.
- **System Preferences**: 49.3% prefer detection when unusual behavior happen ; 25.3% favor continuous monitoring.
- **System Features**: Nicla Vision and DL improve detection accuracy. Arduino-based system provides real-time audio alerts to ensure safety.

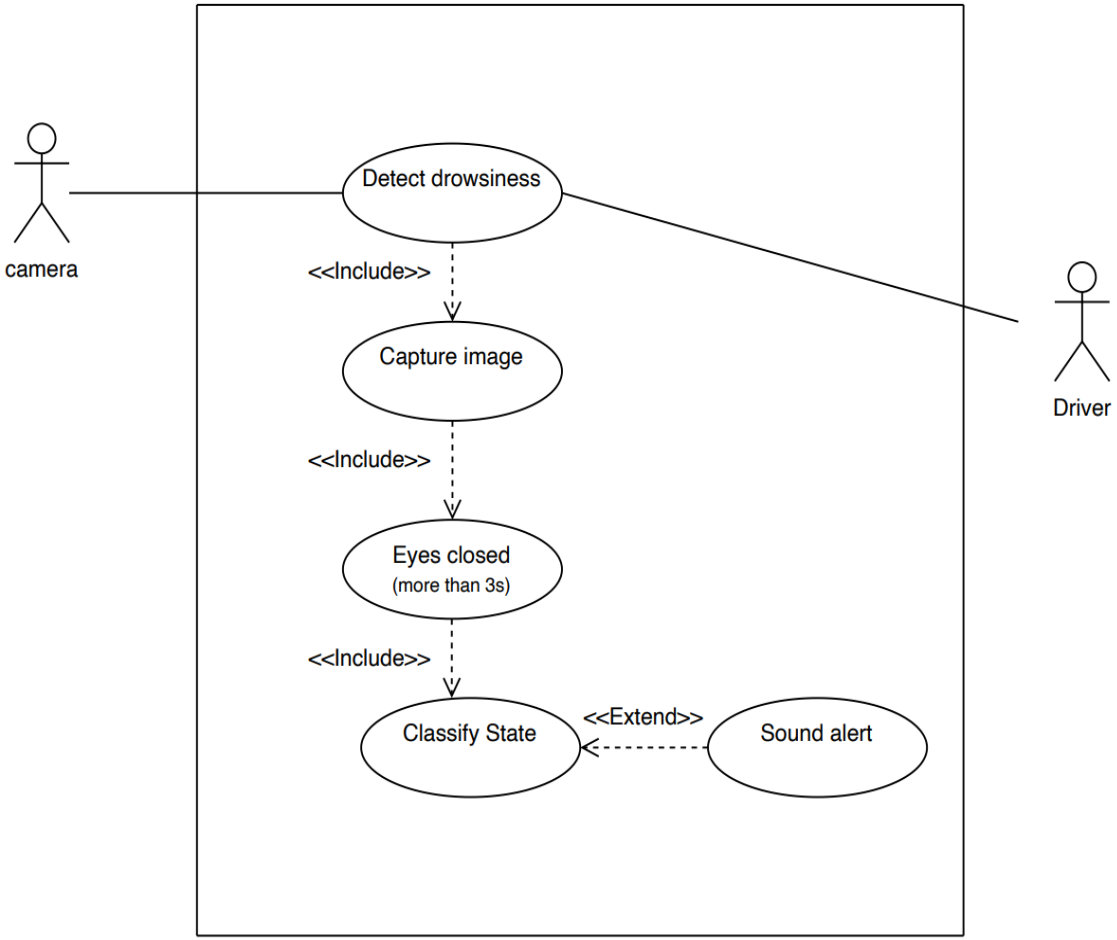




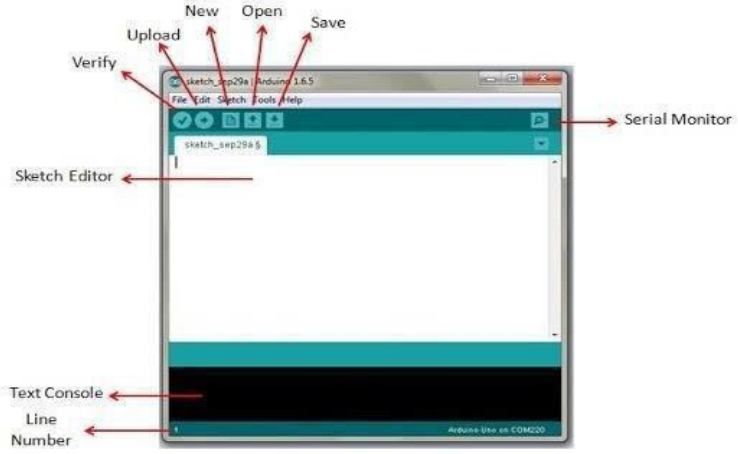


Use case

- **Operational Flow**:
- The system monitors the driver to detect any signs of drowsiness
- The camera captures an image of the driver’s face to analyze eye pattern .
- The system checks if the driver’s eyes are closed for more than 3 seconds.
- The camera Classifies the driver's state as **drowsy** or **not drowsy**.
- **Alert Mechanism** : Generates an alert if drowsiness is detected
- **Driver Response** ot snotica yrassecen ekat revird eht pleh strelA: .ytefas niatniam



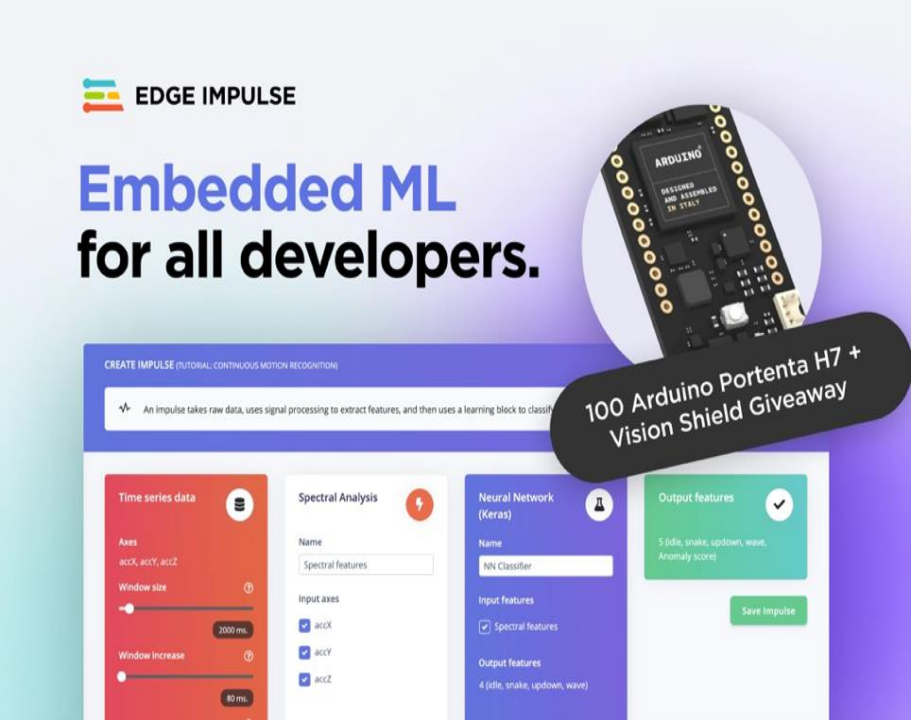
Software Requirements
Arduino IDE



Arduino IDE

- **Overview**:
  - A software platform for writing, compiling, and uploading code to Arduino-compatible hardware.
  - Connects hardware components (e.g., Nicla Vision camera) to the software for real-time image processing of the driver's face.
- **Key Features**:
  - **Simple Editor**: Write, edit, and save code as "sketches".
  - **Toolbars**: Quick actions like save, verify, and upload code.
  - **Message Area & Console**: Displays feedback and logs for debugging.
  - **Board Info**: Shows configured board and serial port for hardware connection.
- **Project Use**: Supports image capture and drowsiness detection with seamless hardware integration.

Software Requirements
Edge Impulse



Edge Impulse

- **Overview**:
  - Platform for developing, training, and deploying machine learning models for edge devices.
- **Role in Project**:
  - Trains the TinyML model to process real-time images from the Nicla Vision camera.
  - Enables high-performance AI tasks on devices with limited computing power.
- **Key Features**:
  - User-friendly interface for uploading data images
  - Facilitates direct deployment of ML models to the Arduino microcontroller.
  - Operates without a constant internet connection.

Hardware Requirements
Arduino Uno

- **Overview**:
  - Popular among beginners for its ease of use and affordability.
  - Uses a USB cable for uploading code, eliminating the need for complex setups.
- **Key Features**:
  - Compatible with devices like LEDs, sensors, motors, and GPS units.
  - Supports interactive project development using the Arduino IDE and programming language.
- **Community Support**:
  - Open-source software fosters innovation and project sharing.
  - Large community contributes code and tutorials.

Arduino Uno

Hardware Requirements
Nicla vision camera

- **Nicla Vision Features**:
  - Compact size and low power consumption.
  - Ideal for rapid prototyping and edge-based image processing.
- **Applications**:
  - Asset tracking.
  - Object recognition.

Nicla vision camera

Functional Requirements

- System should continuously monitor the driver's eye closure.
- System should process the data from the camera and sensors in real-time using the Arduino microcontroller.
- System should be capable of Detecting the driver's eyes, If the eyes remain closed for more than 3 seconds, it will classify the image as indicating a drowsy driver.
- System should trigger an alert (sound and light) when drowsiness is detected.

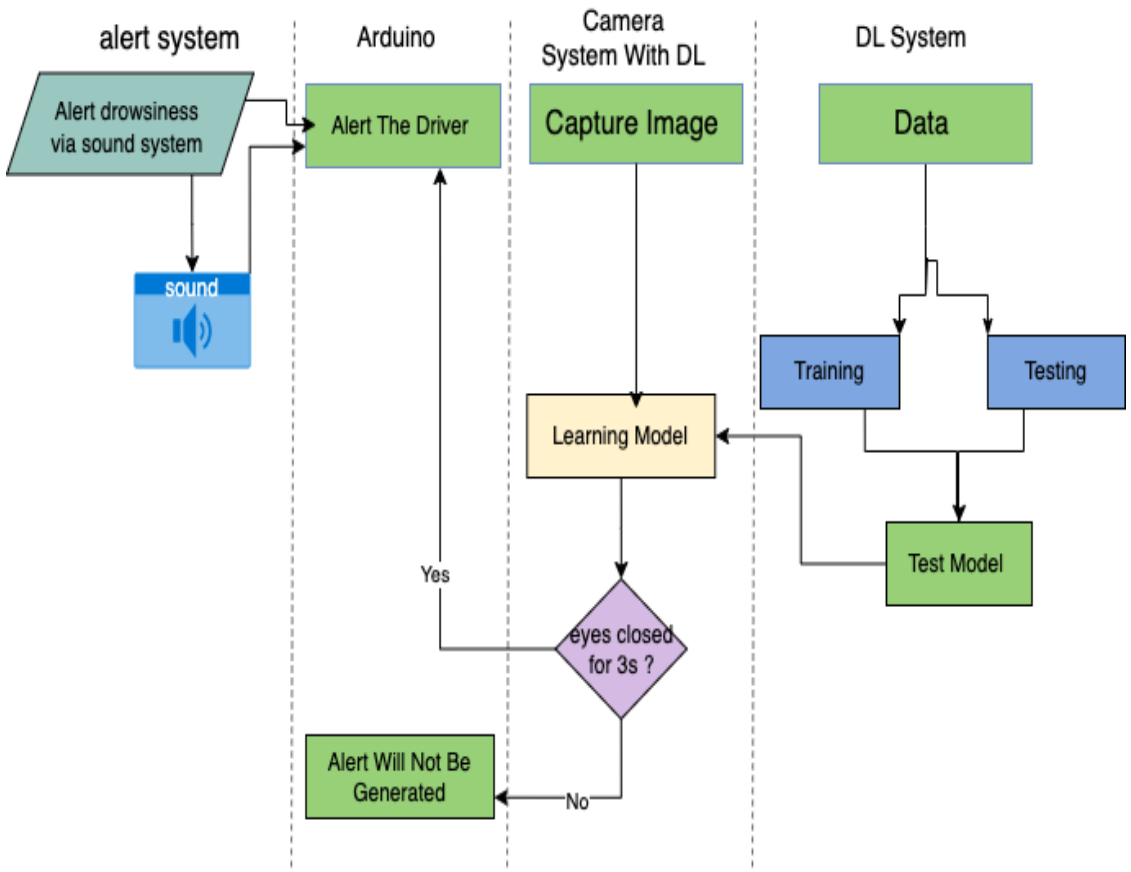
Non-functional Requirements

- **Accuracy:** the system should detect drowsiness with a high level of accuracy.
- **Response time:** the system should be done in a certain period in detecting motion and sending an alert.

System Architecture

- **System Architecture:**
  - **Nicla Vision Camera:** Captures images of the driver's eyes.
  - **AI Algorithms:** Uses AI algorithms to analyze and classify eye closure duration.
  - **Arduino Microcontroller:** Receives classification result from the camera and processes the data.
- **Alert System**:
  - No alert if no drowsiness is detected.
  - If eyes remain closed for over 3 seconds, the system classifies the driver as drowsy.

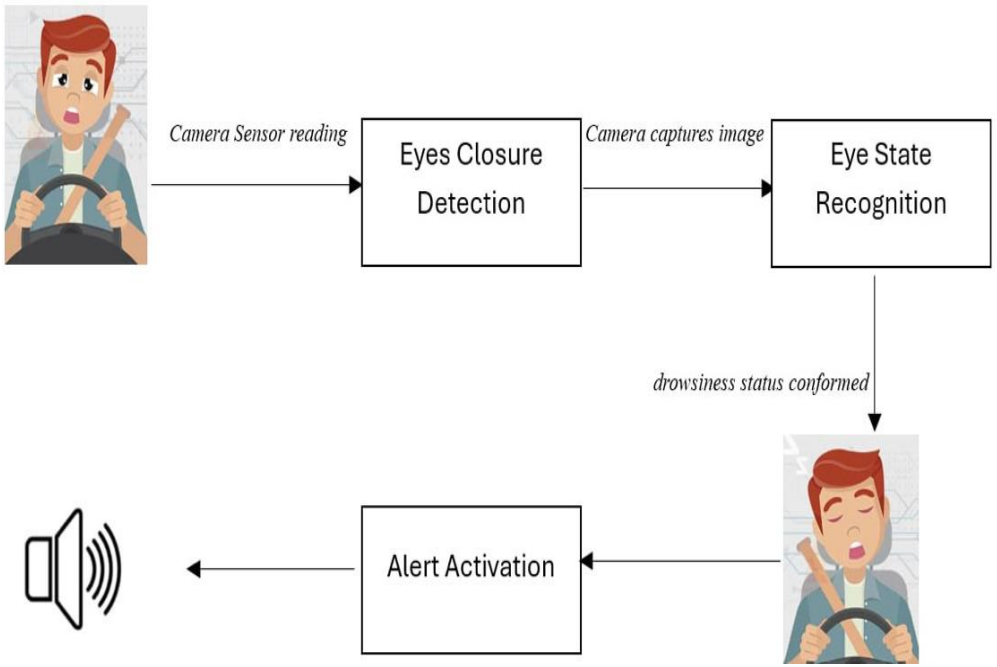
Audio Alert: A buzzer sounds to prompt the driver.



System Prototype Modules & Activity diagram

The system is composed of three primary modules:

- **Eyes Closure Detection:** Uses the Nicla Vision camera and Arduino to detect signs of drowsiness(ex: prolonged eye closure)
- **Eye State Recognition :** Confirms signs of drowsiness by checking eye closure.
- **Alert Activation:** Triggers an audio alert if drowsiness is confirmed .



System Prototype Modules

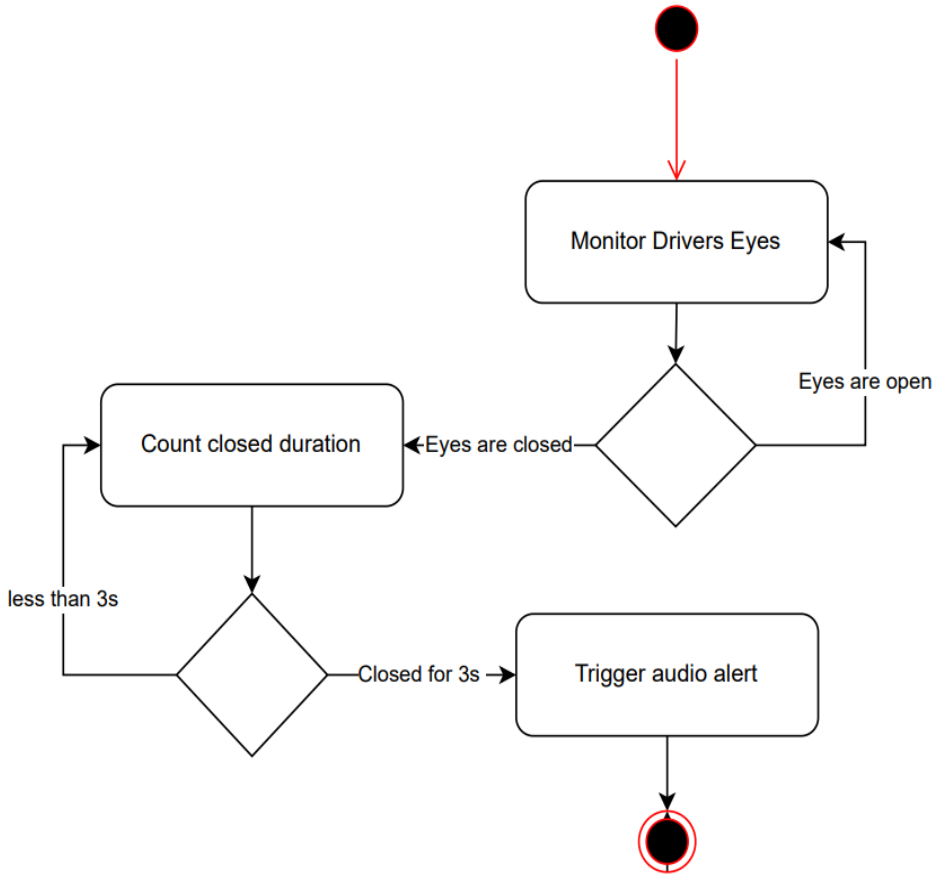
System Prototype Modules & Activity diagram

The flow of the activities in the activity diagram includes the following:

- **Monitor Driver's Eyes:** Continuously track the driver's eyes to check if they are open or closed.
- **If Driver's Eyes closed :** Start counting how long the eyes stay closed.
- **If Open:** Reset the counter and continue monitoring.
- **Duration Check:**

  Less than 3 seconds: If eyes are closed for less than 3 seconds, return to monitoring.

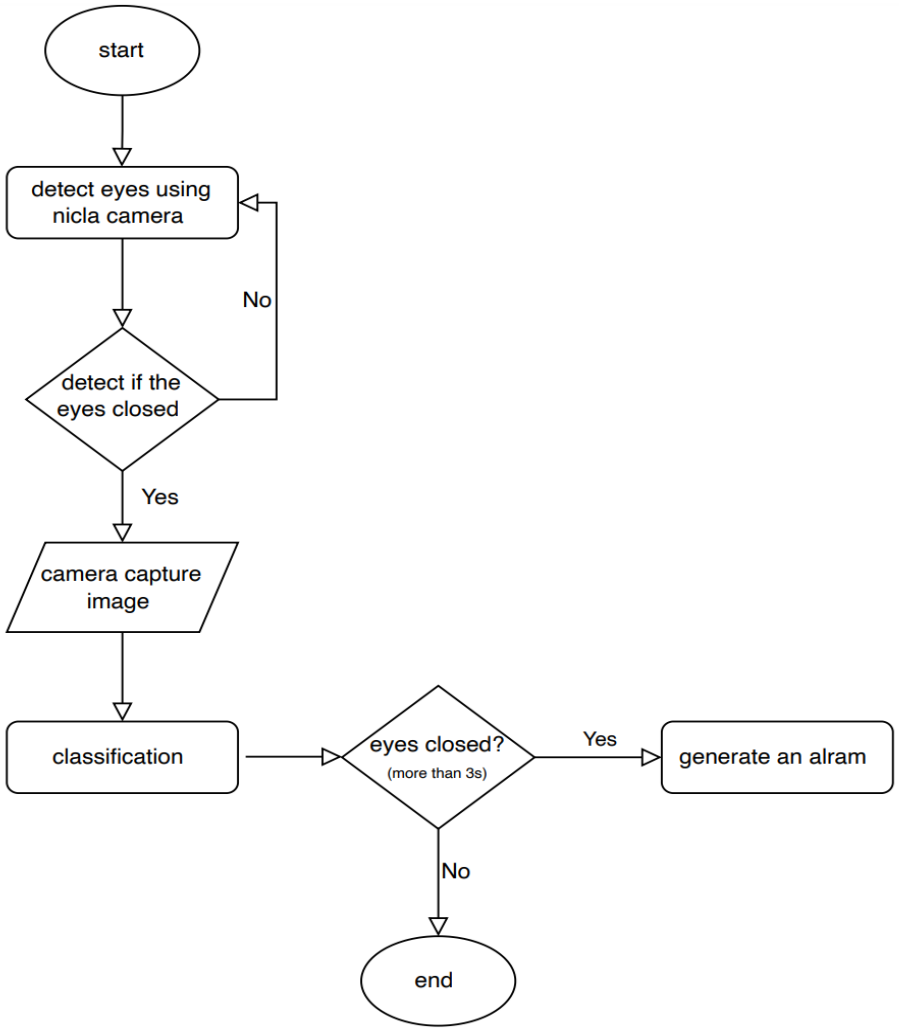
  More than 3 seconds : If eyes remain closed for 3 seconds or more, proceed to trigger alerts.
- **Trigger Alerts:** Activate an audio alert first to warn the driver.



Flowchart

Operational Steps:

- System continuously monitors the driver’s eyes using the Nicla Vision Camera.
- detect if eyes closed
- camera capture image for further analysis
- If eyes are closed for more than 3 seconds
- The system classifies the driver as drowsy .
- If drowsiness is detected an alarm is generated to alerts the driver, else monitoring continues.



User Interface Design

Our project demonstrates the core functionality of detecting driver drowsiness through real-time eye state analysis.

- **Objective**: Develop a hardware prototype to validate the technical feasibility of drowsiness detection.
- **Autonomous Operation**: The system works independently, providing audible alerts without user interaction.
- **No UI Included**: A user interface was excluded to avoid added complexity and keep the focus on the primary goal of detecting drowsiness effectively.

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Thank You